A timing cost of local quantum readout in autonomous noisy dynamics

Research derivation and proof record

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Abstract

Removing an external clock changes more than the order of error-correction operations. A device that must deliver an unknown quantum state at a fixed small port at a chosen late physical time must also generate a sufficiently narrow interval of local quantum availability. We prove a lower bound on that operation for finite-dimensional, bounded-rate, time-independent local Lindblad dynamics, including coherent Hamiltonian terms and all mutable controller registers.

The allowed spatial Pareto burst process contains an exact admissible assignment of memoryless local replacement channels. Under this assignment, local availability of entanglement consumes a finite conditional-information budget. After a local regularization argument, the same noisy evolution has an exact Poisson representation that limits its late-time variation. Combining the two yields $N = \Omega(T)$ installed registers and $U_{\mathrm{all}} = \Omega(T^2)$ register-time for fixed nontrivial quantum fidelity at a fixed compact port at time T, at one fixed positive noise intensity. This holds with known prepared controllers; it does not invoke arbitrary-junk restart.

The theorem excludes carrying the accepted sublinear synchronous hardware bill into this continuous-time model while retaining its public-deadline local-output contract. It does not construct an autonomous fault-tolerant processor, prove a matching upper bound, or exclude output at a causally selected random completion time. The result is a new derivation relative to the supplied record, not an independently reviewed theorem or a claim of literature priority.

1 Question selected and accepted capital

The selected question is:

> Can the accepted sparse processor retain its fixed-deadline, compact-port quantum input/output contract and sublinear hardware when the synchronous tick is replaced by bounded-rate, time-independent local dynamics?

The answer is no within the model below. The reason is independent of the particular clock-repair rule.

The complete sparse proof record and its preceding reusable construction were read at commit `6113f5274d937b751f632a82e03e7f63bf0d234a` [1, 2]. Their accepted theorems are imported without reproof. In ambient $d \geq 2$, write $r = \alpha - d > 1$. At a sufficiently small fixed physical intensity, the synchronous construction acquires an unknown compact input, retains reference correlations, and returns it to a fixed compact raw port at a declared time $t_T = T(1 + o(1))$, with fixed error at most $1/4$, using

$$L = T^{1/r+o(1)}, \qquad N = T^{1/r+o(1)}, \qquad U_{\mathrm{all}} = T^{1+1/r+o(1)}.$$

The actual-source compiler, strong arbitrary-input normalization, noisy controller preparation, sacrificial I/O, sparse live-value execution, two-track birth/disposal identity, fixed-intensity padding construction, and local raw-interface error remain accepted at their stated scopes.

The synchronous tick and patterned horizon-dependent hardware are retained premises of those results. Nothing here invalidates them. The separate arbitrary-junk restart theorem is also preserved, with its distinct startup contract. It is not used in this proof.

2 The operation and continuous-time model

The device D contains N finite-dimensional registers. Its register dimension, spatial density, local interaction range, and number of generator terms per register are bounded independently of the horizon. Its finite layout, initial known controller state, and program may depend on the horizon. All mutable clocks, addresses, programs, quantum data, syndromes, buffers, and I/O records belong to D.

After admission of the unknown input, the ideal law is time-independent:

$$\dot{\rho} = \mathcal{L}_0(\rho), \qquad \mathcal{L}_0 = \sum_{a=1}^{M} \mathcal{L}_a, \qquad M \le m_0 N. \tag{1}$$

Each $\mathcal{L}_a$ is a Lindblad generator on a bounded spatial support. The support Hilbert-space dimension is at most d_*, and its normalized Choi operator has norm $\|\mathsf{J}(\mathcal{L}_a)\|_\infty \le j_*$. These constants are uniform across horizons. This includes local Hamiltonian terms, local dissipative resets, and updates conditioned on corruptible controller registers. It includes the jump-channel form

$$\mathcal{L}_0 = \sum_a \gamma_a(\Phi_a - \mathrm{id}), \qquad \sum_a \gamma_a \le \gamma_* N, \tag{2}$$

used in the continuous-time outline of Dünnweber–Styliaris–Trivedi (DST) [3].

The physical faults retain the supplied law:

$$\lambda\, dx\, dt\, \nu(dR), \qquad \Pr(R \ge u) = u^{-\alpha}, \quad u \ge 1. \tag{3}$$

A mark permits an arbitrary CPTP map on registers inside its ball. Other registers may be left unchanged. A universal guarantee against these faults must hold, in particular, for the memoryless assignment constructed below. Thus no assumption of independent effective decoder faults is made.

Let R_{ref}, abbreviated R, be a qubit reference never acted on. The input is one half of a Bell pair. It is independent of all known apparatus preparation. A fixed compact output subsystem P is read through a fixed CPTP map $\mathcal{R} : P \to O$, with O a qubit. Write $B = D \setminus P$. At a deterministic public time $t_* \ge T$, require

$$\tfrac{1}{2}\|\mathcal{E}_{t_*} - \mathrm{id}\|_\diamond \le \delta < \frac{1}{2}. \tag{4}$$

The accepted target is $\delta = 1/4$. The map $\mathcal{R}$ is allowed to be ideal for the lower bound, so physical local readout errors cannot weaken the conclusion.

One terminal local access at the stated time is allowed. An externally timed command that starts a long decoding or release operation is a different operation: before terminal access, all decoding and release must occur under (1). There is no uncounted noiseless output store or input-dependent restoration source.

The proof even grants an arbitrary initial encoding of the input and arbitrary known apparatus preparation. Granting these advantages is a lower-bound device, not a new positive startup construction. It follows that the bound applies to the original, weaker raw-acquisition and noisy known-controller startup.

3 Main result

Theorem 1 (Late local quantum output requires linear hardware)**.** *Fix the uniform local bounds in (1), a fixed compact output port, and one positive intensity λ. Suppose a family satisfies (4) under every admissible assignment of the burst fault maps. Then, for sufficiently large T,*

$$N(T) \geq cT, \qquad U_{\mathrm{all}}(T) \geq cT^2, \tag{5}$$

where $c > 0$ can depend on the fixed noise law, local bounds, port size, and fidelity target, but not on T.

For the pure jump model (2), the proof gives $N = \Omega(\lambda^2 T)$ when time and length units and all other constants are fixed. For general local Lindblad generators, the displayed regularization gives $N = \Omega(\lambda^3 T)$ at small λ. These powers of noise strength and numerical constants are sufficient converse estimates, not asserted optima. At the fixed positive intensity of the programme, both imply (5).

The theorem holds for identity storage and for a fixed one-qubit unitary output, by composing the mathematical readout with its inverse. A universal unknown-input quantum processor includes identity storage as a subtask.

The sharper abstract statement used in the proof is the following. Suppose an admissible noisy law has both

$$\mathcal{G} = \mathcal{K} + q(\mathcal{D}_P - \mathrm{id}) = \Lambda(\Phi - \mathrm{id}), \tag{6}$$

where $\mathcal{K}$ generates CPTP dynamics, Φ is CPTP, and $\mathcal{D}_P(\rho) = \mathrm{Tr}_P(\rho) \otimes \tau_P$ replaces the port by a fixed state. Set

$$F(t) = \langle \Phi^+ | (\mathrm{id}_R \otimes \mathcal{R}) \rho_{RP}(t) | \Phi^+ \rangle, \qquad g = \frac{1}{2} - \delta > 0. \tag{7}$$

If $F(t_*) \geq \frac{1}{2} + g$ and $t_* \geq g^2/q$, then

$$\boxed{\Lambda \ \geq \ \frac{q^2 g^6}{324(\ln 2)^2}\, t_*.} \tag{8}$$

All entropies below use natural logarithms.

4 Exact assignment from the actual burst source

Lemma 2 (Permitted local replacement substreams)**.** *For every uniformly local device in (1), the actual burst process admits a memoryless CPTP fault assignment whose averaged noise is*

$$q \sum_{a=1}^{M} (\mathcal{D}_{S_a} - \mathrm{id}) + q(\mathcal{D}_P - \mathrm{id}), \tag{9}$$

where $S_a = \mathrm{supp}(\mathcal{L}_a)$, $\mathcal{D}_{S_a}$ is complete depolarization on that support, and $q = c_{\mathrm{src}}\lambda > 0$ is independent of the horizon. A smaller common q can always be chosen.

Proof. Choose a bounded-radius ball of allowed centers around an anchor of each support, and a fixed radius interval $[R_0, R_1]$ large enough that every resulting burst covers that support. The center volume $v > 0$ and radius probability $R_0^{-\alpha} - R_1^{-\alpha} > 0$ are fixed.

The graph connecting overlapping center balls has bounded degree, by the uniform density and bounded local term multiplicity. Color it with a fixed number K of colors. Add an independent finite label to each Poisson mark as part of the randomized fault assignment. A label selects a color; within that color at most one center ball contains its center. If the radius is in the chosen interval, replace

just the corresponding support and leave all other registers unchanged. Otherwise apply the identity. Reserve another label and covering region for P.

These are disjoint measurable sets in the marked Poisson source space. Their restrictions are independent Poisson processes of rates bounded below by a fixed multiple of $\lambda v(R_0^{-\alpha} - R_1^{-\alpha})/K$. Additional identity thinning gives the same rate q for every term. The labels are allowed convex mixtures of CPTP fault maps; they do not change the prescribed physical burst geometry or increase its intensity. Every unused physical mark acts as the identity. □

For the pure jump model, only the port replacement stream is needed. The result uses bounded radii only. It does not rely on the large-radius Pareto tail, on conditioning future noise to be favorable, or on postulating independent coarse faults.

5 A finite budget for local quantum availability

Lemma 3 (Conditional-information drain)**.** *Under the first representation in (6),*

$$q \int_0^\infty I(R : P \mid B)_{\rho(t)}\, dt \le I(R : D)_{\rho(0)} \le 2 \ln 2. \tag{10}$$

The same inequality holds with any finite upper integration limit.

Proof. Apply the port-only semigroup for a small duration h:

$$e^{hq(\mathcal{D}_P - \mathrm{id})} = e^{-qh}\,\mathrm{id} + (1 - e^{-qh})\mathcal{D}_P.$$

The R marginal is fixed. Mutual information $I(R : D)$ is convex over joint states with this fixed marginal, by joint convexity of relative entropy. Replacing P leaves mutual information $I(R : B)$. Consequently this step decreases $I(R : D)$ by at least

$$(1 - e^{-qh})[I(R : PB) - I(R : B)] = (1 - e^{-qh}) I(R : P \mid B).$$

The intervening evolution generated by $\mathcal{K}$ cannot increase $I(R : D)$, by data processing. A Lie–Trotter product, followed by the limit $h \downarrow 0$, proves the finite-time integral inequality. Finite-dimensional entropy is continuous, so this argument also covers states with zero eigenvalues, without assuming an entropy derivative there. Finally $I(R : D) \le 2S(R) \le 2 \ln 2$, and monotone convergence gives (10). □

We next convert (10) into a directly observable fidelity bound. We use the corrected one-way-LOCC theorem of Brandão–Christandl–Yard (BCY), version 5 [4]:

$$I(A : C \mid E)_\rho \ge \frac{1}{8} \big\| \rho_{AC} - \mathrm{SEP}(A : C) \big\|^2_{\mathrm{LOCC}\to} \quad \text{(natural-log units)}. \tag{11}$$

This is imported mathematical machinery.

Lemma 4 (Port fidelity charges information)**.** *For any joint state of RBP and any fixed decoder $P \to O$,*

$$I(R : P \mid B) \ge \frac{2}{9} \left[F - \frac{1}{2} \right]_+^2, \qquad [x]_+ = \max\{x, 0\}. \tag{12}$$

Proof. Decode P locally for the bipartition $R : P$. On R and O, choose uniformly one of the three Pauli tests $XX, -YY, ZZ$ and accept the corresponding positive parity. This is a one-way-LOCC binary measurement with effect

$$M = \frac{1}{2} I + \frac{1}{6}(XX - YY + ZZ) = \frac{1}{3} I + \frac{2}{3} |\Phi^+\rangle\langle\Phi^+|. \tag{13}$$

Every separable state has Bell overlap at most $1/2$. Hence the acceptance probability differs from that of any separable state by at least $(2/3)[F - 1/2]_+$. The binary measurement's outcome ℓ^1 distance is twice this quantity. Composing with the port decoder remains one-way LOCC on $R : P$, so

$$\|\rho_{RP} - \mathrm{SEP}(R : P)\|_{\mathrm{LOCC}_\to} \geq \frac{4}{3}[F - 1/2]_+. \tag{14}$$

Substitution in (11) proves (12). □

Combining (10) and (12) yields the first main ingredient:

$$\boxed{\int_0^\infty \left[F(t) - \frac{1}{2}\right]_+^2 dt \leq \frac{9 \ln 2}{q}.} \tag{15}$$

An unknown input supplies only a finite budget of correlations with its reference. While the output is locally quantum-accessible, a reset of that port irreversibly spends part of this budget. Extra blank ancillas and known clocks cannot replenish it.

This statement is quantum-specific. A classical bit can be copied into the rest of the apparatus so that resetting a readable local copy costs no conditional information. No corresponding classical linear-memory claim follows from (15).

6 Late-time smoothness of time-independent dynamics

Lemma 5 (Poisson smoothness)**.** *If $\mathcal{G} = \Lambda(\Phi - \mathrm{id})$, with Φ CPTP, any fixed effect $0 \leq M \leq I$ evaluated on the evolving state satisfies*

$$|F'(t)| \leq \frac{1}{2}\sqrt{\frac{\Lambda}{t}}, \qquad |F(t) - F(s)| \leq \sqrt{\Lambda}(\sqrt{t} - \sqrt{s}) \quad (0 \leq s \leq t). \tag{16}$$

The initial state and the channel may depend on the chosen device and horizon.

Proof. The exact semigroup identity is

$$e^{t\mathcal{G}} = e^{-\Lambda t} \sum_{k=0}^{\infty} \frac{(\Lambda t)^k}{k!} \Phi^k. \tag{17}$$

Write $F(t) = \sum_k p_k(t) f_k$, where $0 \leq f_k \leq 1$ and $p_k(t)$ is Poisson with mean Λt. The score $p_k'/p_k = k/t - \Lambda$ has mean zero and variance Λ/t. Cauchy–Schwarz gives

$$|F'(t)| \leq \sqrt{\mathrm{Var}(f_k)\Lambda/t} \leq \tfrac{1}{2}\sqrt{\Lambda/t}.$$

Integrate to obtain the second inequality. □

This is the elementary time-estimation/kinetic ingredient, related to existing Markov-clock uncertainty bounds [5]. Its use here is to control a fixed quantum output effect. No clock observable, unbiased estimator, independent clock cycles, or renewal assumption is imposed.

6.1 Why coherent internal dynamics are included

A noiseless Hamiltonian generator generally cannot be written as $\Lambda(\Phi - \mathrm{id})$ at finite Λ. Applying (16) to such a generator without a further argument would be invalid. The arbitrary CPTP burst model supplies precisely the local regularization needed.

Lemma 6 (Local Poisson representation after replacement noise)**.** *Let $\mathcal{L}$ be a Lindblad generator on a Hilbert space of dimension n, set $j = \|\mathsf{J}(\mathcal{L})\|_\infty$, and let $\mathcal{D}(X) = \mathrm{Tr}(X)I/n$. For any $q > 0$,*

$$\mathcal{L} + q(\mathcal{D} - \mathrm{id}) = a(\Psi - \mathrm{id}) \tag{18}$$

for a CPTP map Ψ, where the sufficient rate

$$a = 2(j + q) + \frac{2n^2 j^2}{q} \tag{19}$$

is finite. The support does not grow.

Proof. Use normalized Choi operators, and write $P_0 = |\Omega\rangle\langle\Omega| = \mathsf{J}(\mathrm{id})$ and $Q_0 = I - P_0$. Lindblad generators are conditionally completely positive:

$$Q_0 \mathsf{J}(\mathcal{L}) Q_0 \geq 0. \tag{20}$$

This also follows directly from their Kraus-jump form: the jump terms are positive on this subspace, and the Hamiltonian and anticommutator terms vanish when compressed on both sides.

Set $\Psi = \mathrm{id} + [\mathcal{L} + q(\mathcal{D} - \mathrm{id})]/a$. Relative to $P_0 \oplus Q_0$, the upper-left entry of $\mathsf{J}(\Psi)$ is at least $1 - (j + q)/a \geq 1/2$. Its lower-right block is at least $qQ_0/(an^2)$. Its off-diagonal norm is at most j/a. The Schur-complement correction is therefore at most

$$\frac{n^2 j^2}{qa} \leq \frac{1}{2}. \tag{21}$$

Thus $\mathsf{J}(\Psi) \geq 0$. Trace preservation follows from trace annihilation by the generator. This proves (18). □

Apply this lemma separately to every local term in (1), using the independent admissible replacement streams from Lemma 2. Including the port stream, the exact averaged generator becomes

$$\mathcal{G} = \sum_a a_a(\Psi_a - \mathrm{id}) + q(\mathcal{D}_P - \mathrm{id}) = \Lambda(\Phi - \mathrm{id}), \tag{22}$$

where

$$\Lambda = \sum_a a_a + q \leq m_0 N \left[2(j_* + q) + \frac{2d_*^2 j_*^2}{q} \right] + q. \tag{23}$$

The global Φ is the corresponding convex mixture of local channels and the port replacement. This representation is a proof identity for the same noisy dynamics. It is not a proposed nonlocal hardware operation or a new effective-noise law.

For (2), no regularization of the other terms is needed: $\Lambda \leq \gamma_* N + q$.

As a check on the coherent step, for $\mathcal{L}(X) = -i[\omega Z/2, X]$ on one qubit, the exact condition for $\mathrm{id} + [\mathcal{L} + q(\mathcal{D} - \mathrm{id})]/a$ to be CPTP is

$$a \geq \frac{\omega^2}{q} + \frac{3q}{4}. \tag{24}$$

Thus a noisy coherent clock has a finite Poisson representation, although its noiseless limit does not. Equation (24) is not a claim about an optimal clock or an optimal hardware-noise tradeoff.

7 Proof of the endpoint rate law and hardware theorem

Proof of (8). By (16), whenever the expression on the right is positive,

$$\left[F(s)-\frac{1}{2}\right]_+ \geq \left[g-\sqrt{\Lambda}(\sqrt{t_*}-\sqrt{s})\right]_+ . \tag{25}$$

Because $\Lambda \geq q$ and $t_* \geq g^2/q$, we have $\Lambda t_* \geq g^2$. Integrating the square of the right side from $s=(\sqrt{t_*}-g/\sqrt{\Lambda})^2$ to t_* gives exactly

$$\frac{2g^3}{3}\sqrt{\frac{t_*}{\Lambda}}-\frac{g^4}{6\Lambda} \;\geq\; \frac{g^3}{2}\sqrt{\frac{t_*}{\Lambda}}. \tag{26}$$

Equation (15) bounds this quantity above by $9\ln 2/q$. Rearranging gives $\Lambda \geq q^2 g^6 t_*/[324(\ln 2)^2]$, as claimed. □

Proof of Theorem 1. The guarantee must hold for Lemma 2's exact permitted assignment. Half-diamond error at most δ, tested on a Bell input, implies $F(t_*) \geq 1-\delta = 1/2+g$. Combine (8) with (23). The bracket in (23) is a fixed finite constant, whereas $t_* \geq T$. Hence $N \geq cT$ eventually. For pure jump dynamics use $\Lambda \leq \gamma_* N + q$ instead. At fixed local bounds and small $q \propto \lambda$, these give the stated sufficient λ^3 and λ^2 prefactors. The installed apparatus remains present for at least T, so $U_{\mathrm{all}} = Nt_* \geq cT^2$. □

Equivalently, for the admissible adversary, (15)–(26) show that nonclassical local output cannot remain uniformly strong at late deterministic times if $N = o(T)$. The conclusion does not presume that an error-correction controller stalls. It grants any clock repair, feedback, or internal computation expressible by (1). More explicitly, for $t \geq 1/(4q)$, applying (8) with the actual positive fidelity excess yields

$$F(t) \leq \frac{1}{2}+\left[\frac{324(\ln 2)^2\Lambda}{q^2 t}\right]^{1/6}. \tag{27}$$

The bound is clipped at one. With $\Lambda \leq C_\lambda N + q$, sublinear hardware implies $F(t) \leq 1/2 + o(1)$ at late public times. The decoder can be optimized for each chosen horizon; it only needs to be fixed when comparing nearby times for that same device.

8 Consequences for access, time, and resources

8.1 A strict separation from the accepted synchronous bill

At fixed $r > 1$, $T^{1/r+o(1)} = o(T)$. Thus the accepted sparse hardware scaling and the inherited fixed-deadline compact-port quantum operation cannot both survive this clock removal.

Resource or contract	Accepted synchronous result	Necessary in the present autonomous model
Installed registers	$T^{1/r+o(1)}$, attained	$\Omega(T)$
All-register-time	$T^{1+1/r+o(1)}$, attained	$\Omega(T^2)$
Initialization	Noisy known-controller writes	Same startup is allowed
Output	Fixed compact port, public time	Same output contract
Status	Completed constructive theorem	General converse; matching construction open

At bounded spatial density, $N \le C(L+1)^d$. Therefore the new theorem also implies $L = \Omega(T^{1/d})$. The accepted complete-device erasure argument continues to apply in continuous time to a static device: its reset destroys the input regardless of ideal timing. Consequently

$$L = \Omega\Big(\max\{T^{1/d}, T^{1/r}\}\Big) \quad \text{at fixed intensity.} \tag{28}$$

No matching spatial upper bound is asserted. Register-time is not energy, heat, fabrication work, or physical laboratory performance.

8.2 A general quantum-availability budget

Equation (15) is already independent of how the ideal controller operates. It says that a fixed noisy port cannot carry a fixed positive entanglement-fidelity excess above 1/2 for an arbitrarily long total duration using one once-supplied unknown qubit.

For several reset streams $q_j(\mathcal{D}_{P_j} - \mathrm{id})$, the same Trotter argument yields

$$\sum_j q_j \int_0^\infty I(R : P_j \mid D \setminus P_j)_{\rho(t)}\, dt \le 2S(R). \tag{29}$$

The ports need not be independent. This is a bound on access to the same quantum information, not an assertion of independent fault values. For disjoint or overlapping ports with fixed decoders, (12) gives the corresponding sum of squared fidelity-excess integrals.

8.3 Allowing a larger output region

If the output access region lies in radius $s \ge 1$, a permitted complete replacement stream has rate $q_P \ge c\lambda s^{-r}$. Apply the local regularization streams at rate $q_0 = c_0\lambda$, independent of s, and use q_P in (8). In the large-horizon regime where the resulting lower bound dominates constants, this gives the necessary tradeoffs

$$Ns^{2r} \ge c\lambda^3 T \quad \text{(general local Lindblad dynamics)}, \tag{30}$$

$$Ns^{2r} \ge c\lambda^2 T \quad \text{(pure channel-jump dynamics)}. \tag{31}$$

Here an arbitrary decoder on the whole region is granted. Its actual local implementation and time cost remain additional obligations. These are necessary conditions, not sufficient access constructions or claims of optimality.

8.4 Public deadlines and completion events differ

The semigroup smoothness bound concerns a fixed-time marginal. Conditioning on a locally observed completion event selects a different channel ensemble. It does not obey (25) merely by substitution $t_* = \tau$ for a random stopping time.

Thus the theorem leaves open an event-driven device that delivers after a causally selected completion event and pays for its request, timing record, detector response, and local receiving operation. It also leaves open external triggering of a paid decoder and noncompact encoded readout. None of these changes is constructed here. In particular, they do not automatically resolve noisy clock recovery, exactly-once quantum operation history, or liveness under the full burst process. A completion flag followed by leaving the state at the same raw port until a public pickup time remains subject to the theorem.

9 Relation to prior work and limits

DST's Sections II.2 and VII provide the relevant comparison [3]. Its discrete-time proof uses a specified-set fault-path norm condition. Section VII proposes finite-neighborhood clock repair and time-independent local CPTP jump dynamics, while explicitly leaving the continuous-time threshold proof as an outline. Its discussion of readout uses a final encoded state and top-level decoding. This theorem does not challenge that possibility. The additional requirement here is automatic conversion to a fixed compact raw port at a specified late physical time.

The synchronizer itself is not claimed as new. The previously recorded deadlock of a naive modulo clock is not treated as a general obstruction. Nor does randomly timing the sparse two-track circuit establish its cohort identity or its actual-source recurrence.

The mathematical inputs are conditional-information data processing, the corrected BCY one-way-LOCC bound, the Poisson semigroup identity, and elementary Choi positivity. Existing clock-uncertainty and noisy quantum-metrology results already establish important timing limitations [5, 6]. The programme-level addition here is their explicit compatibility with a quantum-output information budget, the actual burst assignment, local coherent regularization, and the resulting resource separation at the inherited interface. The targeted source search did not establish a literature-priority claim for this combination.

The precise limits are:

- The theorem is a converse for time-independent finite-dimensional local Lindblad dynamics with uniformly bounded local strength and term density. It is not a statement about all possible non-Markovian reservoir models or unbounded-dimensional oscillators.
- Persistent accessible environments and traveling input-bearing records must be counted. A noiseless uncounted output sink changes the contract.
- The fixed port and deterministic-time requirement are essential. An autonomously protected encoded state accessible through a spatially large decoder is a different endpoint.
- The fault guarantee is adversarial over allowed CPTP assignments. The proof chooses a particularly simple Markovian replacement assignment. It does not infer a lower bound for a narrower noise law that forbids these maps.
- The proof does not claim an optimal intensity prefactor, a practical threshold, a matching $N = O(T)$ construction, or a completed autonomous acquisition/computation/delivery protocol.
- The known-controller startup remains distinct from arbitrary-junk restart. Identical powers of T in the two lower bounds do not make their assumptions or mechanisms interchangeable.

10 Finite diagnostics and retained unresolved work

The accompanying `check_readout_bound.py` validates the finite-dimensional identities and inequalities used here:

- 54 local Lindblad/depolarization cases in dimensions $2, 3, 4$: minimum Choi eigenvalue 3.29×10^{-7}, maximum trace-preservation error 7.86×10^{-17}, and maximum generator-identity residual 4.39×10^{-14}.
- 12 qubit cases check the exact boundary (24) from both sides.
- 100 full-rank reference/rest/port states satisfy the local information-drain inequality.

- 18 Poisson score cases satisfy the smoothness bound. The Pauli witness identity (13) is checked directly.

These are diagnostics for algebra and normalization. They do not simulate a protected processor, prove the theorem by sampling, or measure a physical noise threshold. The proof is the argument above.

The positive continuous-time programme remains open. Its remaining central object is a physical finite-state control architecture that repairs corrupted timing, preserves the quantum operation history, makes progress under actual bursts, acquires the raw unknown input, and delivers the output with a complete resource bound. A fixed-time compact-port version must additionally meet (5). An event-driven version must explicitly earn its different output contract.

References

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